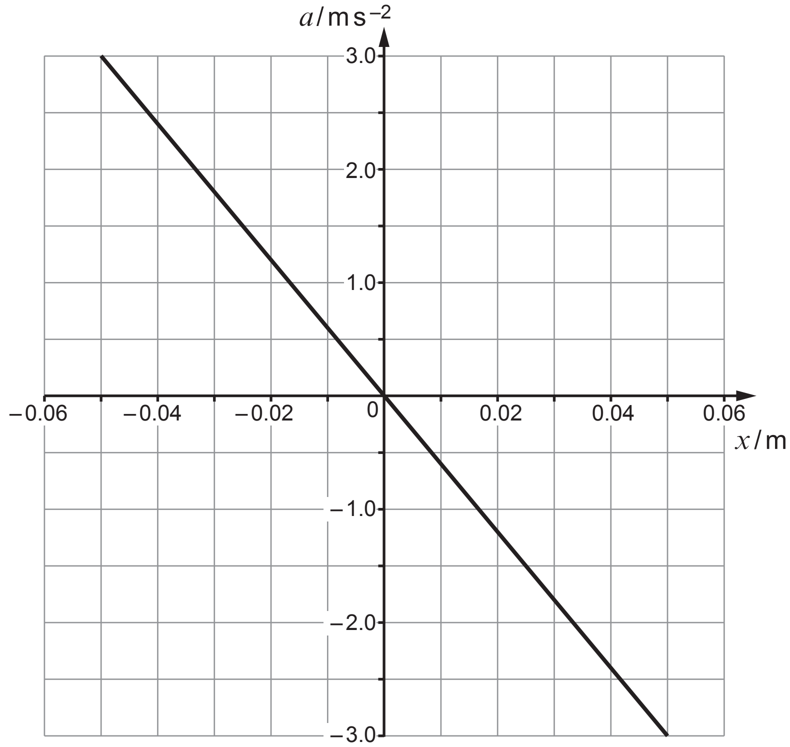
Examiner only

3. A small object of mass 0.30 kg oscillates at the end of a spring. The graph shows how its acceleration, a, depends on its displacement, x, from a fixed point.



(a) (i) Identify the **two** characteristics **of the graph** that show that the object is oscillating with *simple harmonic motion.* [2]

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(ii) Use the graph to show that the angular frequency, ω, of the oscillation is approximately 7.7 rad s^{-1}. [3]

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Examiner only

(iii) Calculate:

I. the period of oscillation; [1]

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II. the spring constant; [2]

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III. the velocity of the object as it passes through the centre of oscillation. [2]

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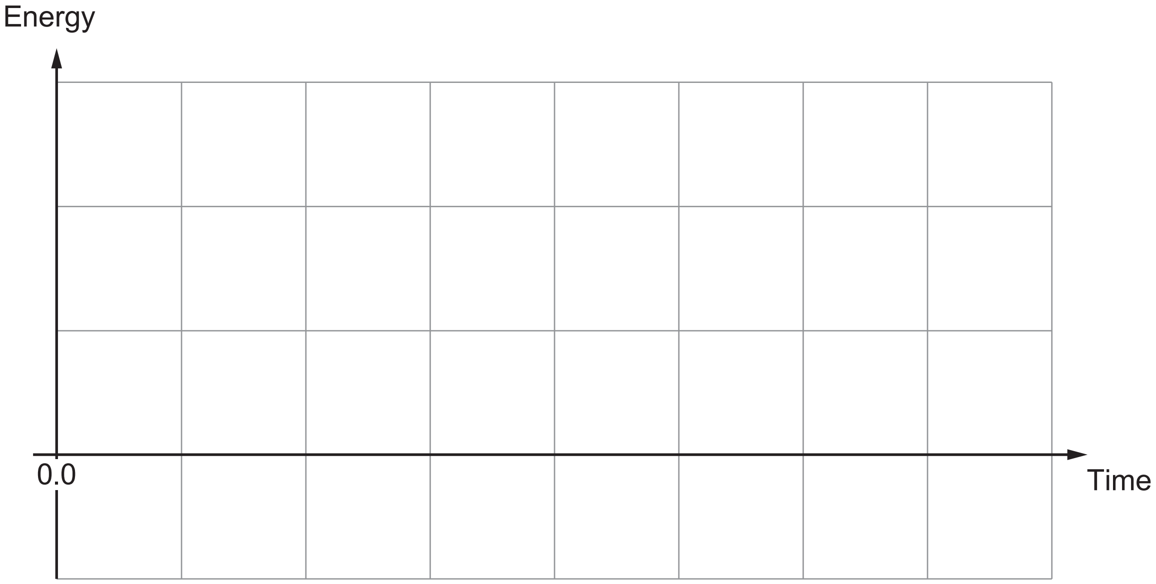
(iv) The object was released at the displacement of –0.05 m at time t = 0. Sketch on the axes below:

I. the potential energy of the system (label the sketch PE),
II. the kinetic energy of the system (label the sketch KE).

Draw both sketches **over one cycle** of the undamped oscillation. (No values are required on the axes.) [3]



Examiner only

(b) Explain what is meant by the terms *damping* and *resonance* **and** discuss their importance in real systems giving an example for each. [6 QER]

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